
Replicating FTS-Diffusion: Degeneracy of the Released Generation Pipeline

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We revisit FTS-Diffusion (ICLR 2024), a financial time-series generator built from
2 scale-invariant subsequence clustering (SISC), pattern-conditioned diffusion, and
3 learned pattern evolution, and try to reproduce it from the released code. In our
4 runs the released sampling pipeline is near-degenerate: synthetic S&P 500 paths
5 are one repeated motif tiled into an almost perfectly linear price ramp (linear-fit
6 $R^2 \approx 1.0$), independent runs are nearly identical, and synthetic return volatility is
7 about $20\times$ too small. We trace this to three choices in the released sampling code:
8 a deterministic argmax rollout of the pattern chain, a one-unit decoder hidden state,
9 and a commented-out magnitude normalization. We are explicit about confounds
10 we cannot rule out: no checkpoints are shipped, and our training budget is reduced.
11 Consistent with degeneracy, a continuous-rollout Train-on-Augmentation-Test-
12 on-Real (TATR) setup appears to improve S&P 500 forecasting by up to 85%,
13 yet the same protocol never beats its baseline on GOOG or ZC=F. Augmentation
14 carrying real structure would help on all three assets; a gain limited to the one
15 trending asset points to a scale coincidence. On the released artifacts the authors'
16 own Kolmogorov–Smirnov and Anderson–Darling return tests cleanly separate
17 synthetic from real in essentially every window, so a simple return check detects
18 the degeneracy that the downstream MAPE misses. We also do not reproduce
19 the Appendix B.3 SISC toy targets. We present these as observations from one
20 reproduction attempt rather than a verdict on the method.

21 1 Introduction

22 Deep generative models are attractive for finance because historical market data are scarce, noisy,
23 nonstationary, and not experimentally reproducible. Financial returns also exhibit heavy tails, weak
24 raw-return autocorrelation, volatility clustering, and recurring shapes at irregular time scales [Cont,
25 2001]. Sequence generators try to preserve temporal structure with adversarial, optimal-transport, and
26 score-based objectives [Yoon et al., 2019, Xu et al., 2020, Tashiro et al., 2021]; financial generators
27 additionally need to respect stylized facts and economically meaningful dependence [Wiese et al.,
28 2020]. FTS-Diffusion [Huang et al., 2024] addresses this by discovering variable-length motifs and
29 generating new trajectories motif by motif.

30 We reproduce FTS-Diffusion and extend its evaluation; we do not propose a new generator. We
31 retrace the released pipeline and try several generation protocols, asking whether any rollout produces
32 non-degenerate samples. We also test whether the SISC toy-data targets reproduce. The architecture
33 is sensible, but in our hands the released sampling pipeline degenerates: the synthetic series collapses
34 to a near-linear price ramp built from one repeated motif (Section 4). Given that collapse, the apparent
35 S&P 500 gain probably reflects a scale coincidence that does not transfer across assets. We cannot
36 fully separate this from a known confound: the original authors release no trained checkpoints, so we
37 train every module ourselves and at a reduced budget relative to their defaults.

Contributions. We make four contributions. (i) A code-level diagnosis of the released sampling pipeline (Section 4): three specific choices that drive the synthetic series to a near-degenerate linear ramp, plus the underspecifications we had to resolve. (ii) A 30-seed S&P 500 TATR study, with the cross-asset observation that the continuous-rollout gain does not transfer to GOOG or corn futures (ZC=F), which we read as a scale coincidence. (iii) A return-distribution check showing that the authors’ own Kolmogorov–Smirnov and Anderson–Darling tests detect the degenerate output that the downstream MAPE still rewards. (iv) A documented Appendix B.3 SISC toy-data gap. Throughout, we separate claims about the released artifacts from what would need the authors’ checkpoints to settle.

2 Background: FTS-Diffusion

FTS-Diffusion decomposes financial time-series generation into three modules [Huang et al., 2024]. Scale-invariant subsequence clustering (SISC) segments a univariate series into variable-length motifs using dynamic time warping (DTW) [Sakoe and Chiba, 1978] and DTW barycenter-style centroid updates [Petitjean et al., 2011]. Each segment is summarized by a pattern identity p_m , a duration scale α_m , and a magnitude scale β_m . A pattern generation module (PGM) then produces concrete segments from these motif conditions; it pairs a scaling autoencoder (Scaling-AE), which encodes and rescales segment shapes, with a pattern-conditioned diffusion model (PCDM) trained on a DDPM-style denoising objective [Ho et al., 2020].

Long-horizon behavior is controlled by the pattern-evolution module (PEM), which learns

$$(p_m, \alpha_m, \beta_m) \mapsto (p_{m+1}, \alpha_{m+1}, \beta_{m+1}), \quad (1)$$

with a classification loss for the next pattern and regression losses for the next duration and magnitude scales. Although useful, this separation makes downstream results sensitive to how the learned state chain is initialized, reset, and divided into synthetic training blocks.

3 Replication protocol

Assets and splits. Our main datasets are daily Yahoo Finance close-price histories for S&P 500, GOOG, and ZC=F. S&P 500 spans 1980–2020 and yields 10,088 trading days and 2,282 post-SISC segments with $K = 14$. GOOG spans 2004–2020 with 3,776 trading days and 733 post-SISC segments with $K = 11$. ZC=F spans 2000–2020 with 4,764 trading days and 629 post-SISC segments with $K = 11$. We follow a chronological train/test design: motif extraction, fitted generator components, standardizers, and downstream predictors are fit only on their allowed training windows, and final scores are computed on held-out real data.

Downstream benchmark. Our main downstream benchmark is TATR: Train on Augmentation, Test on Real. We divide the held-out real slice into an initialization portion and a final real test portion. A one-day-ahead LSTM forecaster is trained on the real initialization portion plus Y synthetic trading years, where one synthetic year has 252 days and Y is swept across a fixed grid. We score with mean absolute percentage error (MAPE) on the final real test portion. We use paper-like LSTM settings: one-day ahead, window size 64, hidden size 32, two layers, MAE training loss, Adam with learning rate 10^{-2} , and 100 epochs for the primary S&P 500 runs.

Synthetic rollout variants. We compare three ways to produce the synthetic augmentation path, each sharing the same data budget. We use the descriptive name with the code identifier in parentheses.

- **Independent fixed blocks** (authors): independent 252-day blocks, each generated from the same fixed initial state.
- **Continuous chunked** (split): one continuous trajectory from the pattern-evolution chain, cut into 252-day chunks for downstream training.
- **Independent blocks with random init** (random_init): independent 252-day blocks, each from a randomly drawn initial segment.

These three protocols differ only in how the Markov chain transitions across blocks. Because the pattern-evolution module is the only component that carries long-horizon state across motifs, block connectivity is what they probe.

86 **Compute.** Our primary 30-seed S&P 500 downstream batch used stored SISC artifacts and PG-
87 M/PEM checkpoints and recorded 10,478 seconds on an Apple M4 laptop with 10 CPU cores and 16
88 GB memory. This time excludes earlier exploratory runs not used for the reported tables.

89 4 Degeneracy of the released generation pipeline

90 We reproduce FTS-Diffusion end to end from the released code. Before downstream scores, we
91 report what the generator actually produces, because it determines how the later numbers should be
92 read. In our runs the released sampling path is near-degenerate: the synthetic price series is a single
93 repeated motif tiled into an almost perfectly straight line. We identify three code choices behind this,
94 with file references into our working copy (fts-diffusion-ref/), then show the degeneracy on
95 the saved trajectories and the return distribution. We treat these as observations about the released
96 artifacts under our setup rather than proofs about the method; Section 4.5 lists the confounds we
97 cannot rule out.

98 4.1 Three choices in the released sampling code

99 **(1) Deterministic argmax rollout of the pattern chain.** FTS-Diffusion rolls out the pattern-
100 evolution module (PEM) by taking `argmax` for the next pattern and the next length, and the raw regres-
101 sion mean for the next magnitude, with no sampling (fts-diffusion-ref/models/sampling.py,
102 lines 30, 33, 34):

```
103 pred_pattern = torch.argmax(probabilities_pattern, dim=1).unsqueeze(0)
104 pred_length = torch.argmax(probabilities_length, dim=1).unsqueeze(0) + 1_min
105 pred_magnitude = pred[:, n_patterns+range_length:].float()
```

106 An autoregressive map $\text{state}_{m+1} = f(\text{state}_m)$ that is deterministic must, iterated from a fixed initial
107 state, converge to a fixed point or a short cycle. Because the paper does not specify the rollout policy
108 (greedy vs. sampling), this greedy reading of the release is defensible, but it removes the stochasticity
109 the Markov-chain description suggests.

110 **(2) One-unit decoder hidden state.** Scaling-AE uses an LSTM decoder whose hidden di-
111 mension is fed from `pgm_params['sae_hidden_dim']`, set to 1 in the released configuration
112 (`model_params.py`; the decoder LSTM is defined in `scaling_autoencoder.py`). With a single
113 hidden unit the decoder has almost no capacity, and we observe it collapsing to a fixed output that
114 ignores both the conditioning pattern and the diffusion latent.

115 **(3) Commented-out magnitude normalization.** A sampling step that would rescale the generated
116 shape to unit range before multiplying by the magnitude β is commented out (`sampling.py`, lines
117 53–55), so the emitted segment has magnitude $\beta \times$ (whatever range the decoder produced) rather
118 than β . In our runs that range is on the order of 0.07, far below the intended scale, which removes the
119 magnitude information β was meant to carry.

122 4.2 What the saved trajectories look like

123 Figure 1 contrasts a real S&P 500 path with one released-pipeline synthetic path. That synthetic
124 series is visually a straight line; a least-squares linear fit gives $R^2 \approx 0.99999$. Three quantitative
125 observations accompany the figure. First, independent runs of a configuration are nearly identical,
126 consistent with a deterministic rollout. Second, the PEM state sequence collapses (Table 1 and
127 Figure 2): on S&P 500 a single pattern is used in 99.6% of segments (a fixed point), while GOOG
128 and ZC=F settle into period-2 cycles. Third, inspecting the trained PGM decoder pattern by pattern,
129 the decoder output is near-constant across all patterns (cross-pattern standard deviation < 0.001): an
130 up-trend pattern and a down-trend pattern yield essentially the same segment. Together these tile one
131 tiny upward-creeping segment across the horizon, producing exactly the linear ramp in Figure 1.

132 4.3 A return-distribution check detects the degeneracy

133 A natural and direct check is the authors’ own return-distribution test. Following their
134 `distribution_tests` routine, we form simple returns and compare real against synthetic with a

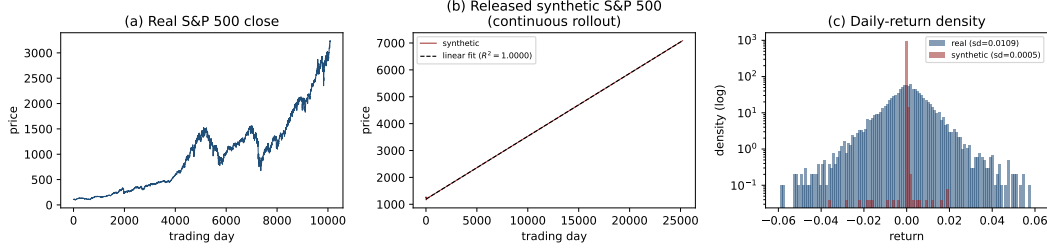


Figure 1: Released-pipeline output on S&P 500. (a) Real close prices show rich structure. (b) One synthetic continuous-rollout path is almost perfectly linear (linear-fit $R^2 \approx 0.99999$). (c) Daily-return density (log scale): synthetic returns concentrate near a single value with standard deviation about $20\times$ smaller than real returns. We show one representative run; runs are nearly identical.

Table 1: PEM state-sequence collapse under the deterministic rollout, from saved `run_*_states.npy`. “Dominant pattern share” is the fraction of generated segments using the most frequent pattern.

Asset (K)	Segments	Dominant pattern share	Regime
S&P 500 (14)	2520	99.6%	fixed point
GOOG (11)	2520	50.0%	period-2 cycle
ZC=F (11)	1867	50.0%	period-2 cycle

135 Kolmogorov–Smirnov (KS) two-sample test and an Anderson–Darling (AD) k -sample test, where a
136 higher p -value means the two return samples are harder to distinguish. Huang et al. report values
137 consistent with indistinguishability (KS around 0.32, AD around 0.12). On the released artifacts
138 the test instead separates the two cleanly (Table 2): across 300 windows per asset and protocol the
139 synthetic and real returns differ in essentially every window, with mean KS p -values at the 10^{-4} to
140 10^{-7} level and no window passing at $p > 0.05$. Return volatility is the mechanism: the released
141 synthetic series has 4 to $60\times$ too little of it, because a near-linear ramp has almost no daily variation.
142 A degenerate generator is expected to fail a distributional test, so failing one is no surprise; the useful
143 contrast is that this simple return check catches the degeneracy while the downstream price-MAPE of
144 Section 5 still rewards the same output.

Table 2: Return-distribution check on the released artifacts. “Mean KS p ” and “% indist.” are over 300 windows of length 60; higher is more similar (the paper reports KS around 0.32). “Real/Syn. sd” are daily-return standard deviations. continuous denotes an un-chunked continuous rollout, authors the independent-block protocol.

Asset	Protocol	Real sd	Syn. sd	Mean KS p	% indist.
S&P 500	authors	0.0109	0.0045	9×10^{-4}	0
S&P 500	continuous	0.0109	0.00046	4×10^{-7}	0
GOOG	authors	0.0184	0.0026	6×10^{-5}	0
GOOG	continuous	0.0184	0.00029	4×10^{-7}	0
ZC=F	authors	0.0181	0.0028	2×10^{-5}	0
ZC=F	continuous	0.0181	0.00041	1×10^{-6}	0

145 4.4 Underspecifications we had to resolve

146 Several choices that matter for the result are not pinned down by the paper. Table 3 lists the ones we
147 encountered, separating the three that are load-bearing for the degeneracy above from the compute-
148 driven confounds and the remaining setup choices. We kept the released values wherever they existed,
149 so that the run reflects the artifact as published rather than our preferences. We intend this as an audit
150 aid rather than an accusation: underspecification is common, but here it coincides with the failure
151 modes.

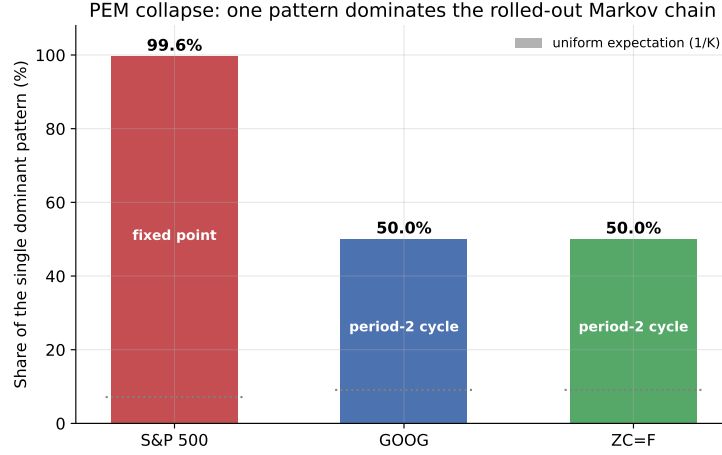


Figure 2: PEM state-sequence collapse under the deterministic rollout. Bars show the share of generated segments that use the single most frequent pattern: 99.6% on S&P 500 (a fixed point) and 50% on GOOG and ZC=F (period-2 cycles). A dashed line marks the uniform expectation $1/K$ that a non-degenerate chain would approach.

Table 3: Paper-vs-release underspecifications we resolved. “Released” marks a value shipped in the released configuration that we kept; “ours” marks a choice we had to make.

Design choice	In paper?	Value used	Note
Scaling-AE decoder hidden dim	no	1 (released)	load-bearing
PEM rollout policy	no	argmax (released)	load-bearing
Magnitude normalization at sampling	implied	commented out (released)	load-bearing
PGM / PEM training epochs	no	30 / 60 (ours; released 400 / 1000)	confound
PCDM diffusion steps	no	30 (ours; released 100)	confound
SISC k -means seed	no	42 (released)	shared seed
GOOG / ZC=F downstream split	S&P only	62.5% train (ours)	cross-asset parity
Initial segment for generation	no	first test segment / random (ours)	protocol switch

152 4.5 Confounds and what we cannot rule out

153 We stress that the evidence above concerns the released artifacts as configured and trained in our
154 hands rather than the method in principle. Two confounds prevent a stronger claim. First, no trained
155 checkpoints are shipped with the release, so every reproduction retrains from the released scripts and
156 we cannot compare against the authors’ own generations. Second, our training budget is reduced
157 relative to the released defaults (PGM 30 vs. 400 epochs, PEM 60 vs. 1000, PCDM 30 vs. 100 steps),
158 a compute-driven choice that a reviewer can reasonably challenge. Three non-exclusive explanations
159 remain that we have not fully separated: undertraining, an unlucky seed, and the one-unit decoder
160 capacity. A full-budget retrain across seeds would falsify or confirm the undertraining hypothesis;
161 we flag it as the natural next experiment and do not assert a method-level conclusion without it.
162 Given how unusual a one-unit decoder hidden state is for a published generative model, it would
163 also be reasonable to ask the authors directly whether that value, and the commented-out magnitude
164 normalization, correspond to the configuration behind the paper’s headline results.

165 5 Replication results

166 5.1 TATR depends on the synthetic-series protocol

167 Huang et al. report that appending 100 synthetic years reduces one-day-ahead MAPE by 17.9%,
168 15.3%, and 17.4% on S&P 500, GOOG, and ZC=F, respectively. Our current S&P 500 evidence
169 has three layers. First, the stored result matrices do not support that trend for the documented

170 authors independent-block protocol: the stored S&P 500 authors curve finishes 208.4% worse
 171 than its baseline, while the continuous-trajectory curve finishes 71.6% better. Second, a one-seed
 172 protocol search found that continuous-rollout variants finish 76.5–85.4% better, whereas independent
 173 fixed blocks finish 96.5% worse. Third, Table 4 reports the primary 30-seed batch across the three
 174 protocols.

175 In that batch, the continuous chunked rollout reproduces a strong paper-like reduction in down-
 176 stream MAPE, while both independent-block protocols worsen it. A single lucky seed does not
 177 drive this: across all 30 seeds the final continuous-chunked change stays below baseline (range
 178 $[-89.1\%, -21.2\%]$), whereas independent fixed blocks range $[+17.0\%, +438.3\%]$ and random-init
 179 blocks $[+3.9\%, +263.3\%]$. Figure 3 shows the same contrast: continuous trajectories cross below
 180 the real-only baseline, while independent blocks worsen as augmentation increases.

Table 4: S&P 500 TATR summary over 30 seeds. Percentages are relative to each protocol’s no-augmentation baseline; lower MAPE is better. “Best change” is the minimum over the augmentation sweep.

Protocol	Baseline	Best change	Final change
Independent fixed blocks (authors)	0.0611	+0.0%	+208.4%
Continuous chunked (split)	0.0620	−77.6%	−72.2%
Random-init blocks (random_init)	0.0611	+0.0%	+68.2%

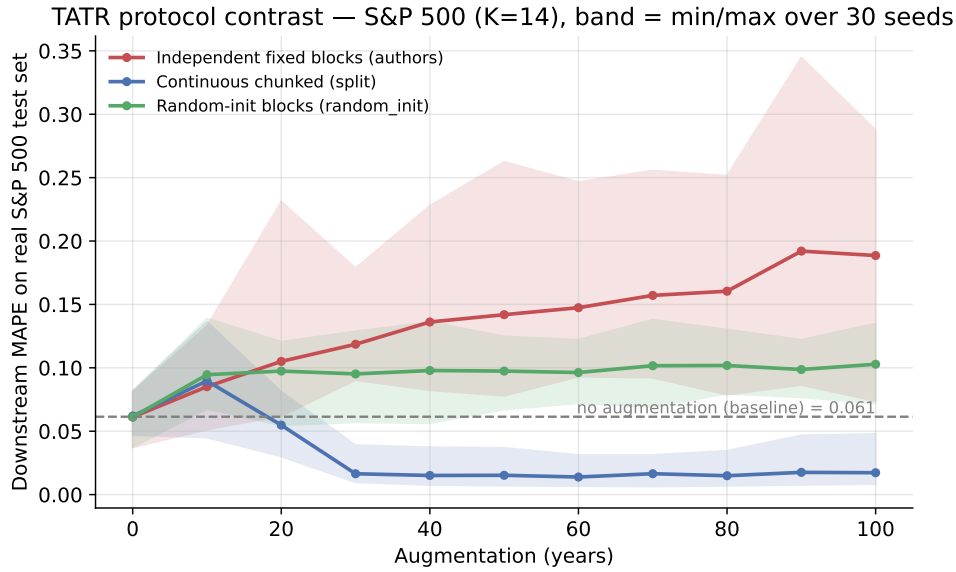


Figure 3: TATR outcome changes sharply with the synthetic-series protocol. Continuous trajectories can look paper-like, while independent blocks worsen with augmentation. A dashed line marks the original S&P 500 100-year TATR target, and shaded bands show min/max across seeds.

181 Synthetic price levels explain the effect. Continuous trajectories start near 1253 and end near 7085
 182 (mean level 4138); independent blocks start from the same level but repeatedly reset, ending near
 183 1249 (mean 1214). Continuous rollouts thus drift the near-linear ramp through the scale of the
 184 upward-trending S&P 500 test period, so percentage error can fall simply because synthetic and real
 185 price levels overlap, not because the dynamics are informative. Independent blocks stay near the
 186 initial level, the overlap never happens, and error grows. Read with Section 4, this looks like a scale
 187 effect rather than useful augmentation. Cross-asset evidence below provides the cleaner test.

188 Train-on-Mixture-Test-on-Real (TMTR) controls, which train the forecaster on a fixed real-plus-
 189 synthetic mixture rather than sweeping the amount of augmentation, make the replication conclusion
 190 more conservative. In our TMTR runs, the reference curve reaches a best mean improvement of
 191 26.8% but finishes 359.7% worse; a continuous offset-30 control reaches a best improvement of

192 50.0% and finishes 17.8% better; a continuous offset-50 control finishes 752.0% worse. So the
 193 evidence does not say that synthetic data always helps. A particular continuous TATR price-level
 194 rollout can create large apparent gains, while other augmentation and evaluation choices do not.

195 We make no claim of misconduct; the available generation details do not pin down the downstream
 196 result uniquely. Once the generator is seen to be degenerate, though, the simplest reading of the
 197 S&P 500 gain is a scale coincidence rather than augmentation that works under the right protocol.
 198 Either way, a reproducible benchmark should treat the Markov-chain rollout, initialization, block
 199 slicing, scaler fitting, and warm-up windows as first-class parameters, and should not rest a positive
 200 conclusion on downstream MAPE for a single trending asset.

201 GOOG and ZC=F test the coincidence reading most directly. We regenerated the stated real-asset
 202 SISC and architecture artifacts for both with $K = 11$, $l_{\min} = 10$, and $l_{\max} = 21$. A strict author-
 203 faithful downstream TATR rerun is blocked by the released S&P-style split logic, which consumes
 204 $252 \times 5 = 1260$ held-out initialization points whereas the GOOG and ZC=F 80/20 held-out windows
 205 contain 744 and 963 points, so we adapted the split while keeping the proportions comparable. This
 206 changes the absolute MAPE level but not the sign of the augmentation effect, since every percentage
 207 is computed within the same split relative to that protocol’s own no-augmentation baseline. Under
 208 that within-split comparison the effect is negative on both assets and for both protocol families.
 209 Independent fixed blocks (authors) finish 1003% worse on GOOG and 75% worse on ZC=F. Nor
 210 does the continuous chunked (`split`) rollout that appears to help on S&P 500 transfer (Figure 4):
 211 its best point over the whole sweep equals the no-augmentation baseline (it never beats it), and it
 212 finishes 92% worse on GOOG and 179% worse on ZC=F. If the generator carried useful structure,
 213 the protocol would not flip sign across assets; a gain confined to the one strongly trending asset points
 214 to a scale coincidence, not augmentation value. We therefore treat GOOG and ZC=F as covered for
 215 SISC/architecture setup and real-asset diagnostics rather than author-faithful downstream replications,
 216 but the within-split sign does not depend on author-faithfulness.

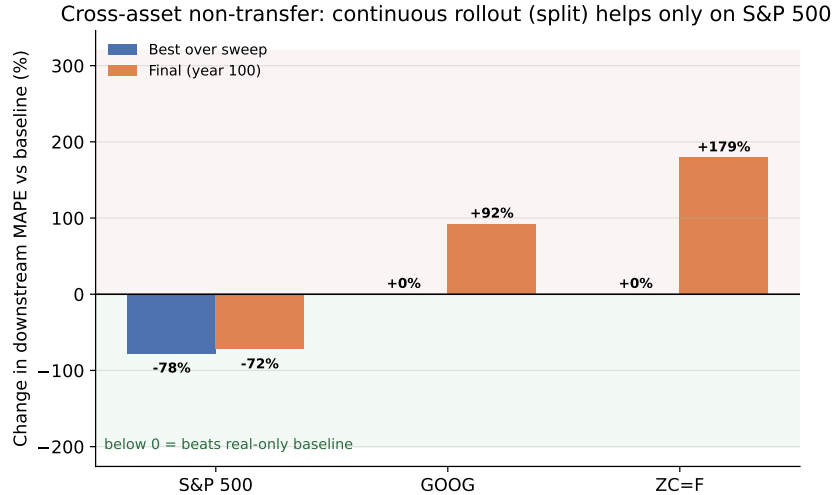


Figure 4: Cross-asset non-transfer of the continuous chunked (`split`) rollout. Bars show the change in downstream MAPE relative to each asset’s no-augmentation baseline (below zero beats the real-only baseline). The rollout helps only on S&P 500 (best -78% , final -72%); on GOOG and ZC=F it never beats baseline and finishes $+92\%$ and $+179\%$. A protocol carrying genuine structure would not flip sign across assets.

217 5.2 SISC toy-data replication gap

218 We also stress-tested SISC on the Appendix B.3 simulated-data setting. Two metrics matter here.
 219 Per-unit DTW is the dynamic-time-warping distance between a recovered pattern centroid and its
 220 target shape, so lower is better. Jaccard is a segmentation-agreement measure, $|A \cap B|/|A \cup B|$,
 221 between the recovered set of segment boundaries A and the ground-truth set B : it equals 1.0 when the
 222 two segmentations coincide and 0 when they share no boundary, and we report a tolerance-2 variant

that counts a recovered boundary as matched if it falls within two time steps of a true one. Huang et al. report one-pattern per-unit DTW/Jaccard of 0.009/0.938 and multi-pattern average per-unit DTW/Jaccard of 0.01/0.784. On our derived one-pattern runs, the best available DTW is 0.0202 and the best tolerance-2 boundary-Jaccard proxy is 0.8693. Across 18 successful multi-pattern runs varying seed, $l_{\max}$, and maximum iterations, our best average per-unit DTW was 0.0321, and our best author-style interval IoU (intersection over union of recovered and true segment intervals) was 0.6418. Figure 5 summarizes the multi-pattern gap.

Appendix B.3 Replication Gap

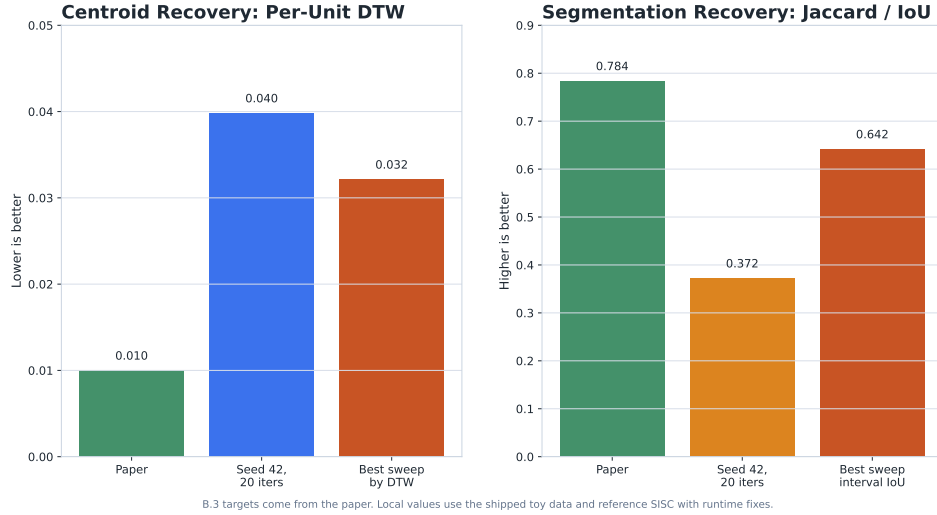


Figure 5: Appendix B.3 SISC replication gap on the toy data. Best replication runs do not reach the reported centroid or segmentation targets.

Increasing SISC iterations from 10 to 20 and changing $l_{\max}$ from 20 to 21 did not close the gap. Our best strict boundary Jaccard at tolerance 2 was 0.3608, and the best reference-code-style interval IoU (intersection over union of recovered and true segment intervals) was 0.6418. Our best centroid metric remained about $3.2\times$ the reported target. We could not reproduce the exact toy benchmark, though this says nothing about the general idea of SISC: the released tree does not expose the exact one-pattern construction, synthetic generator, standard pattern arrays, seed state, or metric implementation used to produce Appendix B.3. More broadly, helper defaults matter: warm-up windows, augmentation scales, plotting conventions, and interval-overlap metrics can all change the apparent result if they go unaudited.

6 Limitations and broader impact

Our replication is constrained by the information and compute available for this study. Most importantly, the degeneracy analysis of Section 4 concerns the released artifacts as configured and as trained in our hands: no checkpoints are shipped, and our training budget is reduced relative to the released defaults. We have not run the full-budget, multi-seed retrain that would separate undertraining from an architectural cause, so we deliberately phrase those findings as observations about the release rather than conclusions about the method. Some runs use reference checkpoints rather than a full fresh retraining of all modules. GOOG and ZC=F have shorter histories than S&P 500, so the downstream split used for S&P 500 is not directly portable without changing the question being asked; we mitigate this by comparing within split, relative to each protocol’s own baseline.

Synthetic financial data can be beneficial for stress testing, augmentation, and reproducibility, but it can also create misleading evidence if artificial predictability is mistaken for real market structure. Any release of synthetic paths should clearly label them as generated data, document the training period and generation protocol, and avoid implying that downstream forecasting gains translate into tradable performance.

7 Conclusion

FTS-Diffusion has a sensible inductive bias for financial time series: variable-length motifs, pattern-conditioned diffusion, and learned motif evolution. Our reproduction attempt, however, did not yield a working generator from the released artifacts. In our hands the released sampling pipeline collapses to a near-linear ramp built from one repeated motif, the authors’ own return-distribution test cleanly flags that output as far from real, and the one downstream setting that appears to improve forecasting does so only on a single strongly trending asset and fails to transfer to GOOG or ZC=F. We read that pattern as a scale coincidence rather than useful augmentation. We do not overstate it: without the authors’ checkpoints and full training budget we cannot yet separate the degeneracy from undertraining, so we report empirical observations about the release, not a verdict on the method. One methodological point holds regardless: realistic-looking paths and a favorable downstream MAPE on a trending series are weak evidence on their own. Future synthetic-finance benchmarks should report exact rollout protocols, direct generator diagnostics such as the return-distribution check used here, and cross-asset transfer, rather than resting positive conclusions on a single asset.

References

- Rama Cont. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.
- Hongbin Huang, Minghua Chen, and Xiao Qiao. Generative learning for financial time series with irregular and scale-invariant patterns. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=CdjnzWsQax>.
- Francois Petitjean, Alain Ketterlin, and Pierre Gancarski. A global averaging method for dynamic time warping, with applications to clustering. *Pattern Recognition*, 44(3):678–693, 2011.
- Hiroaki Sakoe and Seibi Chiba. Dynamic programming algorithm optimization for spoken word recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1):43–49, 1978.
- Yusuke Tashiro, Jiaming Song, Yang Song, and Stefano Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Advances in Neural Information Processing Systems*, volume 34, pages 24804–24816, 2021.
- Magnus Wiese, Robert Knobloch, Ralf Korn, and Peter Kretschmer. Quant GANs: Deep generation of financial time series. *Quantitative Finance*, 20(9):1419–1440, 2020.
- Tianlin Xu, Li K. Wenliang, Michael Munn, and Beatrice Acciaio. COT-GAN: Generating sequential data via causal optimal transport. In *Advances in Neural Information Processing Systems*, volume 33, pages 8798–8809, 2020.
- Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.